Portable Auricular Fibrillation Detector*

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Atrial fibrillation (AF) is a common cardiac arrhythmia that requires regular monitoring. However, traditional 12-lead electrocardiograms (ECG) are hefty, expensive and, more importantly, must be operated by a trained professional. Ultimately, their continuous use is rendered extremely impractical. So as to overcome this problem, we present a portable AF detection device that simplifies the conventional ECG setup to a three-lead configuration. Using the BITalino platform and Python for signal acquisition and processing, respectively, the system provides a reliable cardiac monitoring outside hospital environments. A processing pipeline was implemented, involving resampling and band-pass filtering, in order to condition the signal for analysis. To automate detection a machine learning model was integrated, trained with an AUC of 0.97, enabling AF classification based on the processed ECG segments. While the current prototype is based on offline evaluation, the design anticipates real-time integration, finally bridging the gap between clinical-grade diagnosis and accessible personal monitoring.

I. INTRODUCTION

Cardiovascular diseases account for one of the leading causes of death worldwide, with atrial fibrillation (AF) being one of the most prevalent arrhythmias. Its detection and continuous monitoring is paramount, as untreated AF easily leads to strokes and heart failure. The current benchmark for its diagnosis is the 12-lead electrocardiogram (ECG) managed, in clinical settings. Whilst these devices are highly reliable, they present severe drawbacks: they are bulky, costly, and require precise electrode placement, thereby involving a trained professional. Given that AF requires regular monitoring, which entails frequent hospital visits, these limitations force healthcare systems to deploy a great deal of resources.

To address these challenges, we propose a portable and user-friendly alternative: a simplified ECG device specifically optimized for AF detection. Our design is built around the BITalino platform, which is a biosignal acquisition toolkit. By reducing the original 12-lead setup down to a simple three-lead configuration, accessibility is significantly improved while maintaining diagnostic reliability. The hardware is complemented by a software pipeline designed to process and analyze the acquired signals, and then fed into a machine learning classifier trained to distinguish between normal sinus rhythm and AF episodes. Our model achieved an AUC of 0.97, demonstrating high diagnostic reliability and hence eliminating the need for a trained operator to interpret the results.

Though the current prototype is preliminary due to time constraints, it is designed for future real-time integration in mind, which would enable immediate AF detection and alerting users to seek promt medical attention. Additionally, the hardware is envisioned to be embedded into wearable formats to further enhance its usability.

II. RELATED WORK

The feasibility of using wearable technology paired with machine learning for AF detection is supported by existing research. Lown et al. [4] showed that by combining a wearable heart rate monitor and a machine learning algorithm one can accurately detect AF and transmit ECG data for confirmation. Their algorithm achieved a 100% sensitivity in AF cases, and a specificity of 97.6% in non-AF identification.

Furthermore, the work of Taegyun Jeon et al. [3] outlines the implementation of portable devices for real-time ECG signal analysis. Though it is a more general approach, as it is not specifically designed for AF applications, it aligns with the goal of developing an accessible ECG data collection without compromising the reliability of the results.

III. SYSTEM DESIGN

A. Hardware Design

The core hardware component of our AF detection system is the BITalino biosignal acquisition platform (FIG. 1, namely, the *BITalino (r)evolution Board Kit* [5].

This platform enables ECG acquisition through its dedicated analog front-end, featuring a 0.5-40 Hz bandwidth, 10-bit ADC resolution and 1000 Hz sampling rate. The device also operates on a 3.7V LiPo battery, providing up to 8 hours of continuous monitoring. Due to its lightweight and compact form ($100 \times 65 \times 6$ mm), and its built-in support for ECG acquisition, it is an appropriate choice for developing a portable monitoring device. It must be noted that to achieve clinical-grade results one must take into account the inherent skin-electrode

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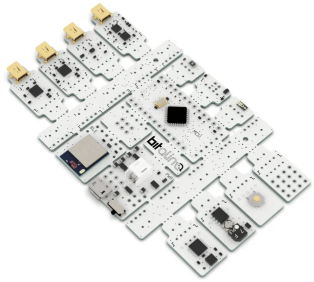

FIG. 1. BITalino system [6]

impedance variations –typically between $1 - 10k\Omega$, and digitally process the signal accordingly.

The ECG module features a three-lead configuration, with lead placement optimized for AF detection, following the specifications of the ECG database used to train the machine learning classifier so as to ensure compatibility between the acquired and reference signals. Following expert consultation[1], we established the following electrode placement protocol: the negative electrode on the right shoulder, while the remaining two are interchangeably placed on the left shoulder and the DII position, as illustrated in FIG. 2.

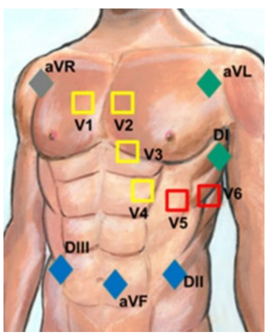

FIG. 2. Electrode placement: *aVR*, *aVL* and *DII* position [1].

In order to enhance usability and guarantee consistent electrode placement, we propose integrating the device into a wearable textile vest. The implementation of said garment is outside the scope of this project, however, a sketch can be found in FIG. 3. This design ensures proper lead spacing, thus eliminating the need for a manual electrode placement by untrained users.

The described hardware configuration, paired with BITalino's Bluetooth data transmission feature, provides the foundation for subsequent digital signal processing, as detailed in the following section.

[1] A. Blanco, personal communication in 2025.

FIG. 3. Prototype of the detection system [2].

B. Software Pipeline

Signal acquisition is eased through the *OpenSignals* software, provided with the BITalino platform. Once a Bluetooth connection is established between the BITalino and the host computer, *OpenSignals* enables real-time visualization and recording of ECG signals.

With the sampling rate configured at 1000 Hz, the recorded data can then be exported in TXT format. These files serve as input to a Python-based processing and classification pipeline, which will thereafter be described. It must be noted that *OpenSignals* supports life information sharing with Python via its API, however, due to time constraints, this feature has not been implemented.

Once imported, the ECG signal is preprocessed so as to standardize and denoise the data prior to feature extraction. The signal is first down-sampled to 250 Hz to match the sampling rate of the training datasheet, thereby ensuring consistent time resolution (note that this process does not generate aliases). A fourth-order Butterworth bandpass filter is applied, with cut-off frequencies set at 0.5 and 40 Hz; this prevents high-frequency aliasing which could resort the signal and lead to errors, while also eliminating lower frequencies which may appear due to the patient's movement. I.e., physiologically relevant data is kept while eliminating noise. Moreover, the European power grid operates at 50 Hz and causes interference, acting as a small capacitance, therefore, in order to suppress powerline interference, a 50 Hz notch filter with a quality factor $Q = 30$ is employed.

Upon filtration, the aim is to retrieve QRS complexes[2], which contain the information needed for AF detection. For that purpose, the GQRS algorithm provided by the WFDB Python library was used, which is a robust toolkit aimed for physiological data analysis. Once

[2] The QRS complex is a part of an ECG that represents the depolarization of the heart's ventricles, leading to their contraction. It typically lasts between 80 to 100 milliseconds and is crucial for diagnosing heart conditions. The time between two successive R waves is the RR interval, used to measure heart rhythm variability.

the QRS complex is detected, RR intervals can be calculated within a 10 second segment.

For each segment, time-domain heart rate variability (HRV) metrics are extracted. Namely, the mean RR, standard deviation of RR intervals, root mean square of successive differences (RMSSD), and the percentage of successive differences exceeding 50 ms (pNN50). These parameters are widely studied to be decisive in AF detection, while enabling potential autonomic real-time implementation.

The extracted features serve as input to the machine learning pipeline based on a Random Forest classifier. The model is trained on labeled segments derived from *PhysioNet* ECG databases, containing both atrial fibrillation and normal sinus rhythm recordings: MIT-BIH Atrial Fibrillation Database and MIT-BIH Normal Sinus Rhythm Database, respectively. The classifier is evaluated using a stratified 5-fold cross-validation: at each iteration de model is trained using 80% of the set, and evaluates the missing 20%. This technique allows for a better assessment of the model's performance and reduces the risk of overfitting.

The trained model is stored and will operate on ECG recordings acquired via BITalino. First, the raw ECG signal is read using the `OpenSignalsReader` module, and the data will be prepared following the same protocol described above. As previously stated, the signal is divided into windows of fixed length. For each window that contains a sufficient number of QRS peaks, the four HRV parameters are calculated. Each resulting vector is reshaped to match the model input.

The classifier outputs a prediction for each segment, indicating whether AF is detected (label 1) or not (label 0). The results for each window are printed to the console, with an indication of how many windows were valid and processed.

All source code and the trained model are available under an open-source license at https://github.com/consigliere0/PEF2-Projecte.git.

IV. RESULTS

To validate the pipeline in a realistic setting, we applied the trained classifier to a raw ECG signal recorded with the BITalino system. The acquisition was carried out using a three-lead configuration and exported in TXT format via the *OpenSignals* software. The recorded signal consisted of 47,250 samples at a 1000 Hz sampling rate, which was downsampled to 250 Hz to match the training conditions. After filtering, QRS complexes were successfully detected; in this particular test case, 64 R-peaks were identified, enabling the extraction of RR interval features.

The filtered signal was segmented into four consecutive 10-second windows. Each window yielded a valid HRV feature vector based on RR statistics, consisting of: meanRR, stdRR, RMSSD, and pNN50. The computed features were:

$$\begin{bmatrix} 0.747 & 0.051 & 0.051 & 50.00 \\ 0.757 & 0.027 & 0.036 & 16.67 \\ 0.703 & 0.038 & 0.033 & 15.38 \\ 0.718 & 0.069 & 0.037 & 15.38 \end{bmatrix}$$

These feature vectors each in turn passed to the trained Random Forest classifier. The predicted labels were:

$$\text{Prediction} = [0, 0, 0, 0]$$

indicating that none of the analyzed segments showed signs of atrial fibrillation. This is consistent with the known ground truth for the subject, who exhibited a normal sinus rhythm during acquisition.

The console output of the pipeline confirmed successful processing:

```
<class 'numpy.ndarray'>
(47250,)
(11813,)
(47250,)
(11813,)
(11813,)
Senyal filtrada i índexs de R detectats.
(64,)
Processed: 1/4 finestres vàlides
Processed: 2/4 finestres vàlides
Processed: 3/4 finestres vàlides
Processed: 4/4 finestres vàlides
(array([...]), array([0, 0, 0, 0]))
```

As stated, the pipeline had been previously trained on labeled ECG segments extracted from the PhysioNet MIT-BIH databases. Classification performance was evaluated using stratified 5-fold cross-validation. Overall, an area under the ROC curve (AUC)[3] of 0.97 was achieved, indicating that the model reliably distinguishes AF from non-AF cases.

These results demonstrate that the model can be applied in practice to signals recorded outside of clinical environments, while retaining classification accuracy.

V. FUTURE DEVELOPMENT

Despite the promising results achieved, this project is still a prototype and thus requires further development before reaching its intended users.

At the moment, the designed system loads an ECG signal, processes it, and it assigns a diagnostic label (either

[3] The area under the receiver operating characteristic curve (AUC-ROC) quantifies the ability of a classifier to distinguish between classes–in this case, atrial fibrillation and normal rhythm. An AUC of 1.0 denotes perfect discrimination, whereas 0.5 reflects random guessing.

0 or 1). Adding a real-time alert system that could warn users when an AF episode is detected, indicating that medical attention is needed, would significantly improve its utility.

Moreover, the current interface should be improved. Given that the intended user is not of technical background, it is important to have a user-friendly interface. For instance, the final output instead of it being binary labels (0 and 1), it should explicitly say whether the patient has AF or not.

A major enhancement would be the integration of real-time signal acquisition and analysis. Currently, the pipeline operates on exported files, which limits its responsiveness. However, the BITalino platform provides a Python-compatible API capable of streaming live data. This would allow the classifier to operate continuously, providing warnings in real time.

As for hardware improvements, the ultimate goal would be to have an entirely wearable design, as described in previous sections. Both the BITalino and the electrode patches would be integrated into the vest, enormously improving its usage experience, and finally bridging the gap between clinical environments and everyday life.

In summary, while the core components of acquisition, signal processing and classification are functional, there is yet work to be done regarding user experience. Such a design would allow for continuous cardiac monitoring in a non-intrusive manner, enabling early detection of AF without requiring medical expertise or specialized infrastructure.

VI. CONCLUSION

This project demonstrates the viability of simplifying the traditional 12-lead ECG based atrial fibrillation detection into a portable and user-friendly design. Through the signal acquisition employing the BITalino platform and the data processing pipeline, we were able to extract relevant heart rate variability features and accurately classify them using a machine learning model. The Random Forest classifier, built with *PhysioNet* datasets, reached an AUC of 0.97, showing that it can reliably distinguish between normal sinus rhythm and AF.

Beyond offline classification, this adaptable pipeline provides a sound basis for future integration into real-time monitoring. With further development, the presented system could autonomously detect AF in daily life, reducing the reliance on healthcare resources.

While challenges remain, this work serves as a relevant step toward more accessible monitoring, empowering individuals to take an active role in their cardiovascular health.

[1] CampusCardio. Las derivaciones en el electrocardiograma. 2018.

[2] Shivam Dixit and Rahul Kala. Early detection of heart diseases using a low-cost compact ecg sensor. *Multimedia Tools and Applications*, 80:1–23, 09 2021.

[3] Taegyun Jeon, Byoungho Kim, Moongu Jeon, and Byung-Geun Lee. Implementation of a portable device for real-time ecg signal analysis. *Biomedical engineering online*, 13:1–13, 2014.

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[5] PLUX wireless biosignals. *BITalino (r)evolution Board Kit Datasheet.* PLUX wireless biosignals, S.A., March 2023.

[6] Plux Signals. Bitalino. 1.